

# INDIAN INSTITUTE OF TECHNOLOGY JODHPUR

## CAPSTONE PROJECT — FINAL REPORT



# Environmental Intelligence & Sustainability Analytics Platform

*A Machine Learning–Driven Environmental Risk Assessment,  
Forecasting & Executive Decision Support System*

*Submitted by*

**Team / Group 14**

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*GitHub: <https://github.com/VahidaSS/CapstoneProject>*

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## Executive Summary

This report documents the six-week capstone project undertaken by Team 14 for the Ecosphere Institute problem statement on Environmental Data Analytics. The project delivers a production-ready Environmental Intelligence & Sustainability Analytics Platform that unifies climate, air quality, and forest sustainability data into a single analytical framework, and exposes the results through interactive Power BI dashboards for executive decision support.

Three heterogeneous data sources — NASA GISTEMP global temperature anomalies (1880–2026), the India AQI air-quality dataset (29,531 records covering PM<sub>2.5</sub>, PM<sub>10</sub>, NO<sub>2</sub>, CO, SO<sub>2</sub> and O<sub>3</sub>), and the Global Forest Watch / FAO forest-cover dataset (2005–2017) — are ingested, cleaned, integrated, and transformed through an automated Python pipeline that persists outputs into both a CSV-based data lake and a PostgreSQL warehouse. Roughly thirty engineered features (rolling averages, trend, growth, lag, cumulative, ratio, and composite indices such as CSI, PEI and FSI) feed two supervised machine-learning models.

The final Random Forest Regressor achieves an  $R^2$  of 0.9925, RMSE of 0.0347, and MAE of 0.0289 on temperature-anomaly forecasting, out-performing the Linear Regression baseline. The platform reports an overall Environmental Health Index (EHI) of 60.18 (Moderate), a Risk Score of 39.82, a Sustainability Score of 60.18, and a Forest Sustainability sub-score of 99.55. Scenario forecasting for 2027–2031 shows a projected temperature anomaly of 0.67 °C under Aggressive-Green policies versus 0.73 °C under Pollution-Growth — evidence that data-driven sustainability action can measurably bend the warming curve.

All code, notebooks, engineered datasets, Power BI files, and phase reports are version-controlled and are available at the project GitHub repository listed on the cover page.

### Headline Results

| KPI                              | Value           | Interpretation                   |
|----------------------------------|-----------------|----------------------------------|
| Environmental Health Index (EHI) | 60.18           | Moderate ecosystem health        |
| Environmental Risk Score         | 39.82           | Moderate — actionable via policy |
| Sustainability Score             | 60.18           | Good                             |
| Forest Sustainability (FSI)      | 99.55           | Excellent                        |
| Climate Score                    | 67.86           | Stable / resilient               |
| Best Model — $R^2$               | 0.9925          | Random Forest Regressor          |
| Best Model — RMSE / MAE          | 0.0347 / 0.0289 | High predictive accuracy         |

## 1. Project Overview and Objectives

### 1.1 Problem Statement

The Ecosphere Institute has commissioned a six-week environmental data-engineering capstone to demonstrate how modern AI and data-science tooling can turn very large, heterogeneous environmental datasets — satellite records, ground-station sensor streams and geospatial land-cover surveys — into actionable insights for climate and pollution policy. Team 14 was tasked with simulating a real environmental data-engineering team: acquiring public data, building an automated ingestion and transformation pipeline, performing analytical modelling and forecasting, and delivering an executive-ready decision-support layer.

### 1.2 Objectives

- Acquire and integrate three heterogeneous environmental datasets (temperature, air quality, forest cover).
- Design a scalable, production-ready ETL pipeline with structured storage in a data lake and PostgreSQL warehouse.
- Engineer analytical features (rolling averages, trend, growth, lag, ratios, composite indices).
- Train and evaluate machine-learning models to forecast climate and pollution trajectories.
- Quantify overall environmental health, risk and sustainability through composite indicators.
- Deliver an interactive Power BI dashboard suite for executive-level decision support.

### 1.3 Scope

The scope covers the full analytical lifecycle: raw data acquisition, cleaning and transformation, automated pipeline engineering, feature engineering, exploratory and inferential analytics, supervised machine learning, scenario forecasting, clustering-based risk segmentation and business-intelligence visualization. The geographic scope is global (temperature and forest data) with a national deep-dive for air quality (India AQI dataset).

### 1.4 Team Composition

| Name                  | Roll Number | Institute Email      |
|-----------------------|-------------|----------------------|
| Vijibenita Ravi Nadar | G25AI1067   | g25ai1067@iitj.ac.in |
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## 2. Datasets and Data Sources

Three complementary public datasets underpin the platform. Together they capture the temperature dimension of climate change, the pollution dimension of environmental health, and the biosphere dimension of forest sustainability.

| Dataset                                            | Source                    | Coverage                        | Key Attributes                                                           |
|----------------------------------------------------|---------------------------|---------------------------------|--------------------------------------------------------------------------|
| <b>NASA GISTEMP — Global Temperature Anomalies</b> | NASA / NOAA               | 1880 – 2026 (monthly & yearly)  | Annual mean anomaly (°C) relative to 1951–1980 baseline                  |
| <b>India AQI Dataset</b>                           | OpenAQ / EPA              | 29,531 city-level records       | AQI, PM2.5, PM10, NO <sub>2</sub> , CO, SO <sub>2</sub> , O <sub>3</sub> |
| <b>Global Forest Cover</b>                         | Global Forest Watch / FAO | 2005 – 2017 (global land cover) | Forest cover (%), geospatial polygons                                    |

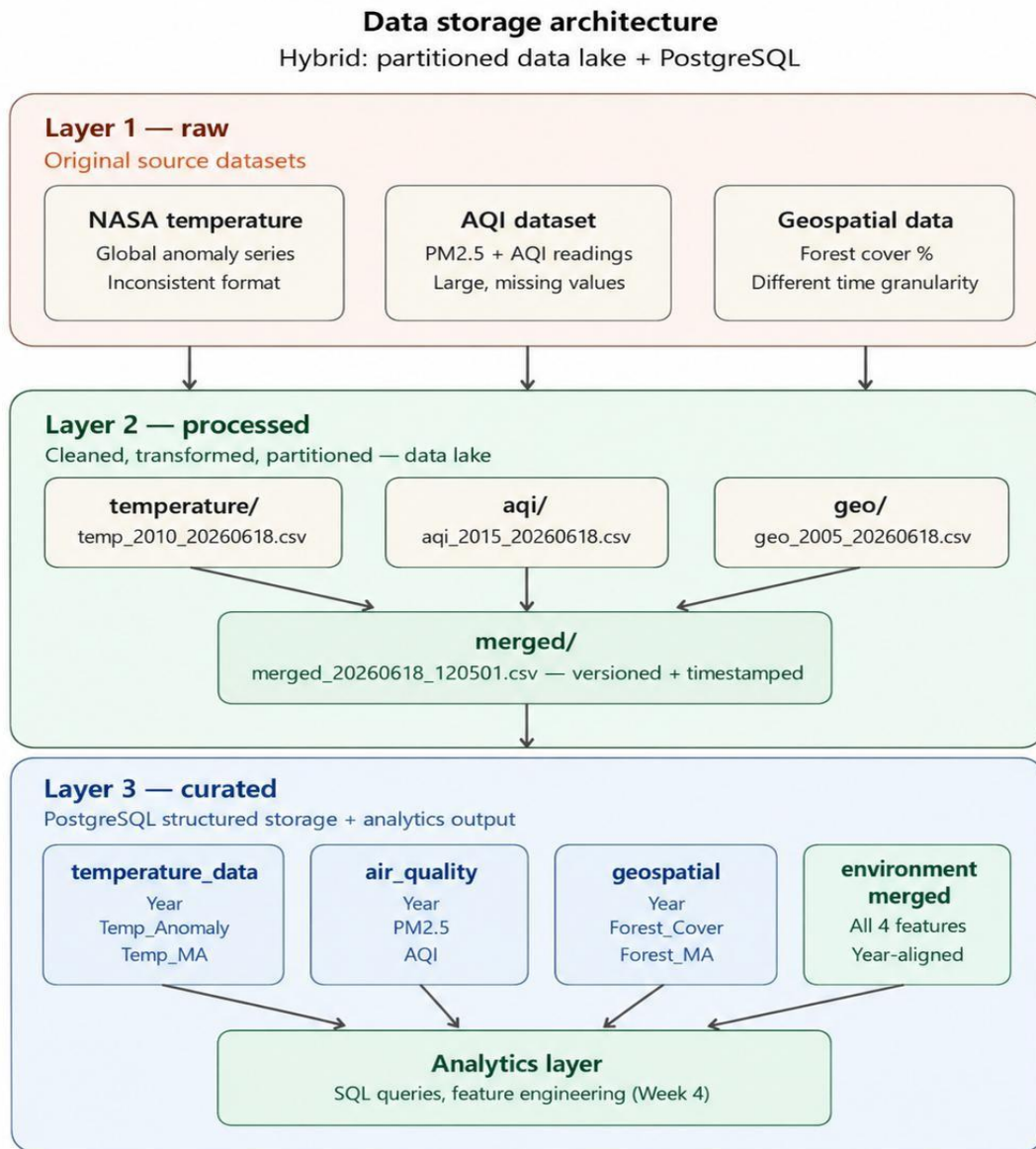


Figure 1: Data Storage Architecture

## 2.1 Data Overlap and Alignment

Because the three datasets have different temporal footprints, the yearly time index (Year) is used as the common integration key. After temporal alignment, only the years 2005–2017 provide simultaneous coverage across all three sources — a material limitation that shaped the design of Week 4 and Week 5.

## 3. System Architecture

The platform is a modern layered data-engineering architecture. Raw environmental data are ingested from public sources, cleaned and validated in Python, feature-engineered into an analytical dataset, modelled with supervised machine-learning algorithms, evaluated against standard regression metrics,



and finally exposed through interactive Power BI dashboards for executive decision-making and sustainability insight.

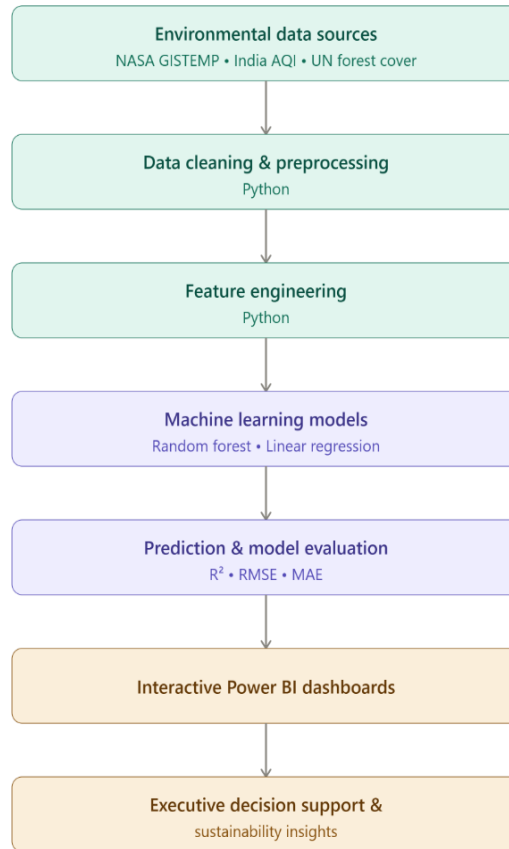


Figure 2: End-to-end platform architecture

### 3.1 Technology Stack

| Layer                | Tools / Technologies                                    |
|----------------------|---------------------------------------------------------|
| Language & runtime   | Python 3.11                                             |
| Data manipulation    | pandas, NumPy                                           |
| Machine learning     | scikit-learn (Linear Regression, Random Forest, KMeans) |
| Storage — data lake  | CSV, partitioned by year and dataset                    |
| Storage — warehouse  | PostgreSQL via SQLAlchemy                               |
| Configuration        | config.json (paths, DB credentials, thresholds)         |
| Logging & monitoring | Python logging → pipeline.log; execution_report.txt     |
| Visualization        | Power BI                                                |
| Version control      | Git / GitHub                                            |

### 3.2 Data Storage Architecture

The solution implements a hybrid storage architecture:

**CSV Data Lake:** Raw and processed data stored in organized directory structure under `/data/` and `/output/` folders. Enables reproducibility and portability without database dependencies.

**PostgreSQL Warehouse:** Structured relational storage via SQLAlchemy ORM. The `environment_merged` schema integrates all three datasets on the `Year` primary key. Enables SQL-based querying for dashboard integration.

**Pipeline Logging:** Automated `pipeline.log` captures ingestion timestamps, record counts, schema validation results, and error flags for operational monitoring.

### 3.3 GitHub Repository Structure

```
CapstoneProject/
├── Architecture Diagram/
│   ├── Data_pipeline_architecture_Week3.png
│   ├── Data_storage_architecture.png
│   ├── environmental_ml_pipeline.png
│   └── ...
├── data/
│   ├── Air_Quality_Data.csv
│   ├── Temperature_Data.csv
│   └── Geospatial_Data.csv
├── misc/
│   ├── Week1_EDA.ipynb
│   ├── Week2_DataCleaning_Structure.ipynb
│   ├── Week3_Capstone_Project_Report.pdf
│   ├── Week4_Capstone_Report.docx
│   ├── Week5Report.pdf
│   └── Week6Report.pdf
├── output/
│   ├── analytics/
│   ├── aqi/
│   ├── charts/
│   ├── dashboard/
│   ├── geo/
│   └── logs/
```

```

|   ├── merged/
|   ├── metadata/
|   ├── plots/
|   ├── reports/
|   └── temperature/
└── powerbi/
    └── CapstoneProjectDashboard.pbix
└── Presentation/
    └── CapstoneProject_Presentation_Team14.ppt
└── Report/
    └── CapstoneProject_FinalReport_Team14.pdf
└── Visualization Dashboard/
    ├── Executive Command Center.png
    ├── Environmental Trends.png
    ├── Climate Change Impact Dashboard.png
    ├── Environmental Health Dashboard.png
    ├── Future Climate Scenario Analysis.png
    └── Predictive Analytics & Machine Learning.png
└── pipeline.py
└── feature_engineering.py
└── analysis_forecasting.py
└── week5_feature_engineering.py
└── config.json
└── feature_description.txt
└── requirements.txt
└── README.md
└── .gitignore

```

## 4 Week 1 — Data Collection & Initial Exploration

### 4.1 Objectives

Week 1 established the analytical foundation of the project. The team identified authoritative public data sources for global temperature, city-level air quality, and forest cover; downloaded the raw datasets; and performed initial exploratory data analysis (EDA) to understand structure, coverage, and quality.

### 4.2 Datasets Selected

- 4.2.1 Temperature — NASA GISTEMP:** global temperature anomaly dataset, monthly and yearly since 1880; structured time-series.

- 4.2.2 Air Quality — India AQI: city-level pollution records including PM2.5, PM10, NO<sub>2</sub>, CO, SO<sub>2</sub>, O<sub>3</sub>, with derived AQI values.
- 4.2.3 Forest Cover — Global Forest Watch / FAO: annual land-cover series (2005–2017) with geospatial extent.

### 4.3 Exploratory Findings

- 4.3.1 Global mean temperature anomaly shows a clear upward trend post-1980, with 2015–2024 the warmest decade on record.
- 4.3.2 The AQI dataset exhibits pronounced seasonal cycles and city-level heterogeneity; PM2.5 is the dominant risk pollutant.
- 4.3.3 Forest cover is nearly flat globally over 2005–2017 but hides sharp regional losses masked by the aggregated series.
- 4.3.4 Data quality issues include missing values (particularly in early AQI records), inconsistent date formats, and mismatched temporal granularity across sources.

### 4.4 Deliverables

- 4.4.1 Data-source inventory (source, license, coverage, granularity, size).
- 4.4.2 Raw dataset copies stored under the /data/raw/ directory.
- 4.4.3 EDA notebook with distribution plots, missing-value maps and time-series overviews.
- 4.4.4 Week 1 progress report.

## 5 Week 2 — Data Cleaning & Transformation

### 5.1 Objectives

Week 2 converted the raw, unstructured environmental datasets into a consistent, analysis-ready format that supports downstream integration and predictive modelling. Cleaning and transformation were engineered to be reproducible and scripted rather than manual.

### 5.2 Tasks Completed

- 5.2.1 Handled missing values through median imputation and rule-based removal of clearly invalid records.
- 5.2.2 Standardized date formats and derived Year and Month features from the raw time stamps.
- 5.2.3 Aggregated the AQI dataset from daily observations into monthly and yearly summaries per city.
- 5.2.4 Filtered and processed geospatial forest-cover data to produce a clean annual series.
- 5.2.5 Applied a moving-average smoother to reduce noise in the temperature and AQI series.
- 5.2.6 Standardized column names, data types, and physical units across the three sources.
- 5.2.7 Aligned datasets on the common Year key and merged them into a unified analytical file.

### 5.3 Outputs

- 5.3.1 cleaned\_temperature.csv, cleaned\_aqi.csv, cleaned\_forest.csv
- 5.3.2 integrated\_dataset.csv (5-feature merged environmental dataset)
- 5.3.3 Correlation heatmap of pollutants, temperature-trend plot, PM2.5 vs. AQI scatter plot, forest-cover trend plot.
- 5.3.4 Week 2 progress report documenting techniques and quality metrics.

### 5.4 Observations & Limitations

Post-alignment, the fully overlapping window across all three datasets is short (only a handful of years contain complete observations). This structural limitation guided later design decisions in Week 4 — specifically, the decision to skip statistically unreliable PCA and correlation analyses on the merged frame in favour of engineered composite indicators.

## 6 Week 3 — Automated Data Pipeline & Storage Architecture

### 6.1 Objectives

Week 3 converted the Week 1–2 exploratory work into a scalable, production-grade data pipeline capable of end-to-end ingestion, cleaning, transformation, aggregation, partitioning, storage, and monitoring — with no manual intervention.

### 6.2 Pipeline Architecture

*Raw Data → Ingestion → Cleaning → Transformation → Aggregation → Partitioning → Data Lake (CSV) → PostgreSQL → Analytics Ready*

The pipeline is modularized into reusable Python components, each responsible for a single concern, and orchestrated by a top-level `run_pipeline()` entry point.

| Function                           | Responsibility                                     |
|------------------------------------|----------------------------------------------------|
| <code>load_data()</code>           | Ingest raw datasets from configured input paths.   |
| <code>process_temperature()</code> | Clean and transform the NASA GISTEMP series.       |
| <code>process_aqi()</code>         | Aggregate AQI data from daily to monthly / yearly. |
| <code>process_geo()</code>         | Clean geospatial forest-cover series.              |
| <code>merge_data()</code>          | Integrate the three datasets on Year.              |
| <code>save_to_postgres()</code>    | Bulk-insert into PostgreSQL via SQLAlchemy.        |
| <code>run_pipeline()</code>        | Orchestrate the end-to-end run.                    |

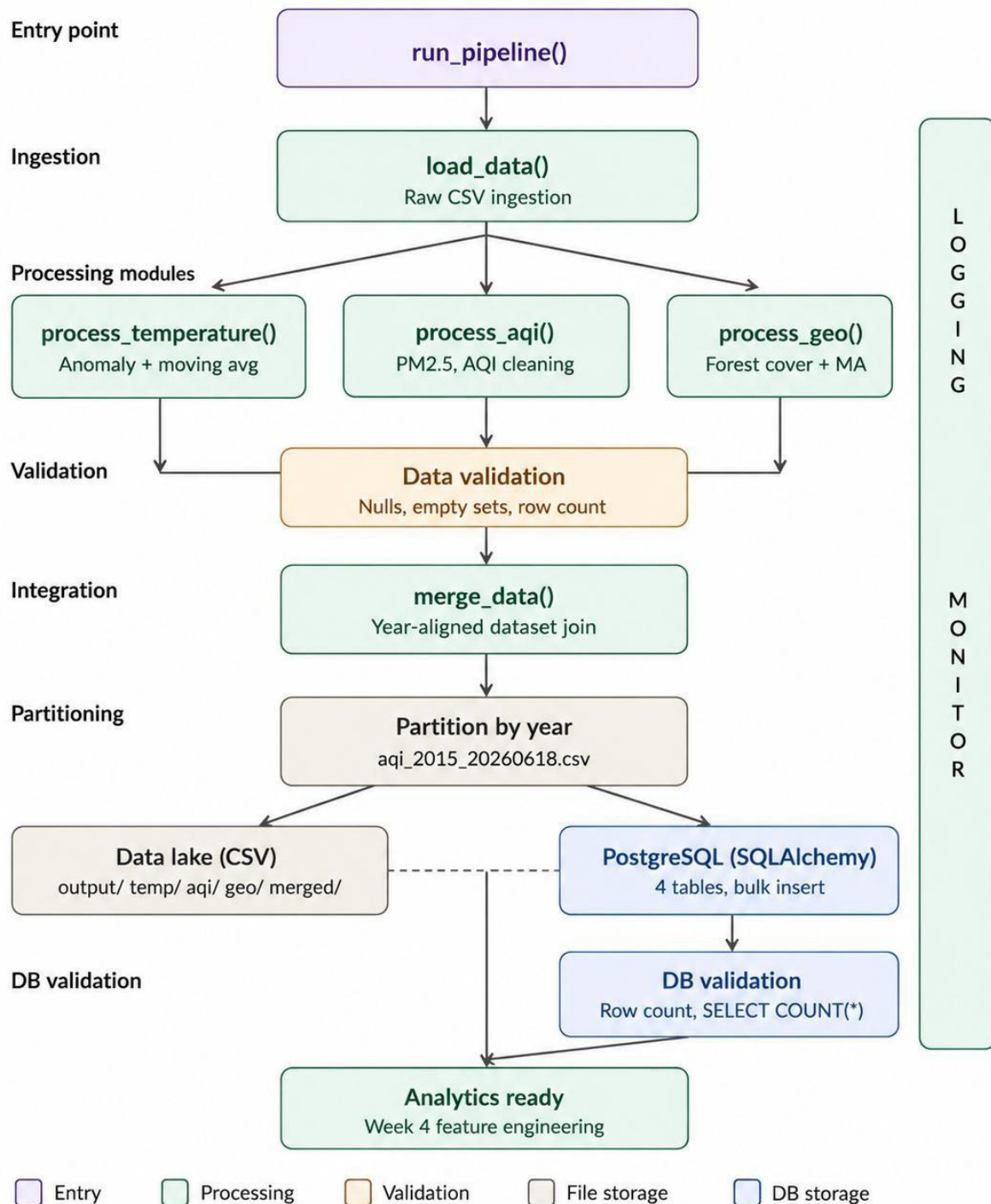


Figure 3: Automated Data Pipeline

## 6.3 Configuration & Monitoring

- 6.3.1 config.json centralizes input/output paths and PostgreSQL credentials — no hard-coded values.
- 6.3.2 pipeline.log records execution steps, per-dataset processing status, warnings and errors.
- 6.3.3 execution\_report.txt captures per-run metadata (record counts, row counts per dataset, and final merged dataset size).

- 6.3.4 Data-quality validation checks flag empty datasets, missing values and schema drift before persistence.

## 6.4 Storage Architecture (Hybrid)

Two storage backends are used in parallel. A file-based data lake gives cheap, traceable historical storage; a PostgreSQL warehouse gives analytics-ready, query table structure.

- 6.4.1 Data lake: Partitioned by Year (e.g. aqi\_2015\_20260618.csv) with output timestamps for full traceability.
- 6.4.2 PostgreSQL: SQLAlchemy bulk-loads four tables — temperature\_data, air\_quality, geospatial, and environment\_merged.
- 6.4.3 Post-insert row-count verification confirms end-to-end integrity for every run.

## 6.5 environment\_merged Schema

| Column              | Type    | Description                           |
|---------------------|---------|---------------------------------------|
| Year                | INTEGER | Common temporal key across sources.   |
| Temperature_Anomaly | FLOAT   | Global mean temperature anomaly (°C). |
| PM2.5               | FLOAT   | Annual mean PM2.5 concentration.      |
| AQI                 | FLOAT   | Annual mean Air Quality Index.        |
| Forest_Cover        | FLOAT   | Annual global forest cover (%).       |

## 6.6 Outcomes

Week 3 marks the project's transition from manual, notebook-driven processing to a validated, monitored, production-grade pipeline. The same pipeline is reused without modification in Weeks 4–6 as the upstream data engine.



## 7 Week 4 — Feature Engineering & Data Transformation Pipeline

### 7.1 Objectives

Week 4 built an automated feature-engineering pipeline that transforms the integrated environmental dataset produced in Week 3 into a rich analytical dataset. Around thirty new features were generated across seven categories, expanding the original 5-column merged frame into an analytical dataset with roughly forty variables.

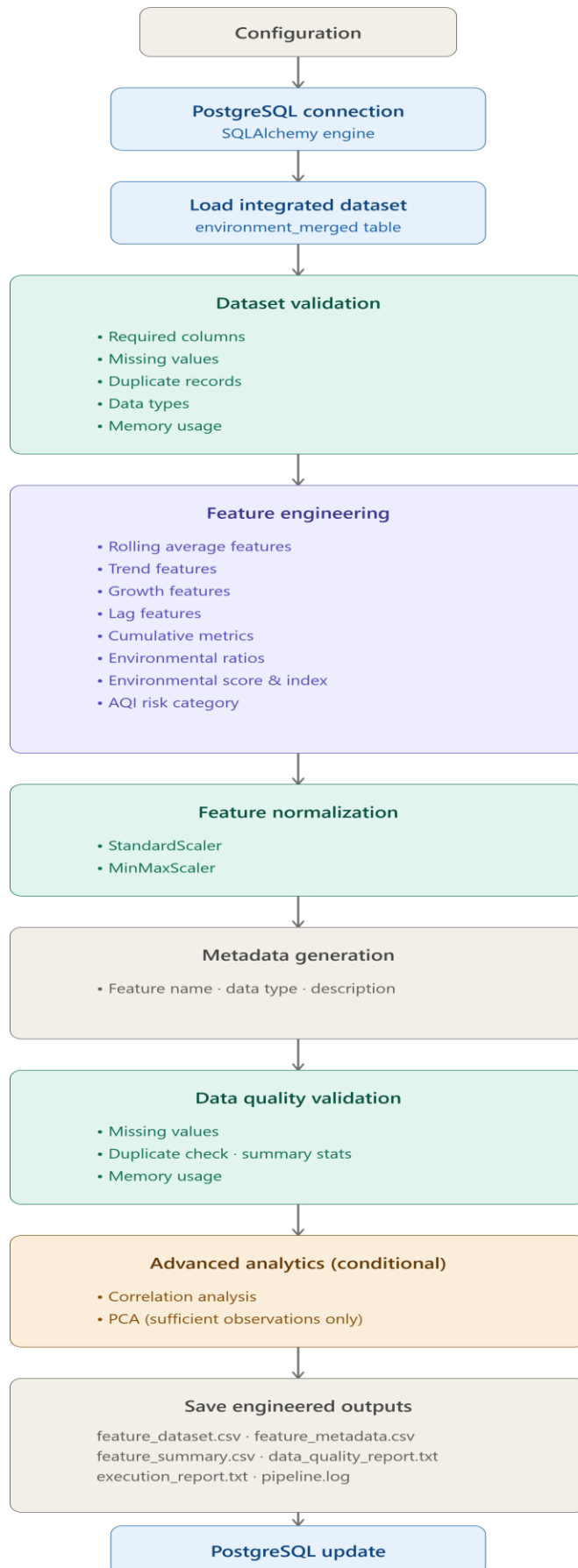


Figure 4: Feature-engineering pipeline

## 7.2 Feature Categories

| Category             | Example Features                                                           |
|----------------------|----------------------------------------------------------------------------|
| Rolling averages     | Temp_5yr_Avg, AQI_3yr_Avg, Forest_3yr_Avg                                  |
| Trend features       | Temp_Trend, AQI_Trend, Forest_Trend                                        |
| Growth features      | Temp_Growth_Pct, AQI_Growth_Pct                                            |
| Lag features         | Temp_Previous_Year, AQI_Previous_Year                                      |
| Cumulative metrics   | Cumulative_Temp, Cumulative_PM2.5, Cumulative_AQI, Cumulative_Forest_Cover |
| Environmental ratios | AQI ÷ Temperature, PM2.5 ÷ AQI                                             |
| Composite indicators | Environmental Score, Environmental Index, AQI Risk Category                |

## 7.3 Composite Indices

Three domain-specific composite indices summarize stress across the climate, pollution and biosphere dimensions:

- 7.3.1 Climate Severity Index (CSI) — measures relative climate stress from temperature anomalies:

$$CSI = (T - T_{min}) / (T_{max} - T_{min}) \times 100$$

- 7.3.2 Pollution Exposure Index (PEI) — weighted composite of pollution exposure:

$$PEI = 0.5 \times PM2.5 + 0.3 \times PM10 + 0.2 \times AQI$$

- 7.3.3 Forest Sustainability Index (FSI) — normalized forest-cover retention:

$$FSI = (Forest\_Cover / Max\_Forest\_Cover) \times 100$$

## 7.4 Normalization

- 7.4.1 StandardScaler produces z-scored variants (mean 0, sd 1):

Temperature\_Anomaly\_ZScore, AQI\_ZScore, PM2.5\_ZScore, Forest\_Cover\_ZScore.

- 7.4.2 MinMaxScaler produces [0, 1] variants for dashboards and composite indicators: Temperature\_Anomaly\_MinMax, AQI\_MinMax, PM2.5\_MinMax, Forest\_Cover\_MinMax.

## 7.5 Metadata and Reproducibility

The pipeline auto-generates feature\_metadata.csv, capturing feature name, data type and description for every engineered variable. Together with the config-driven design and detailed pipeline.log, this ensures the entire feature layer is reproducible from raw data.

## 7.6 PCA & Correlation Note

During execution, the pipeline correctly detected that the strictly-aligned merged environmental dataset

contained too few overlapping observations to support statistically valid PCA or correlation analysis.

Consequently, these steps were intentionally skipped in favour of the engineered composite indicators — an example of the pipeline's built-in guard-rails.

## 7.7 Outputs

- 7.7.1 feature\_dataset.csv, feature\_metadata.csv, feature\_summary.csv
- 7.7.2 data\_quality\_report.txt, execution\_report.txt, pipeline.log
- 7.7.3 PostgreSQL table feature\_engineered with database validation report
- 7.7.4 Source code: feature\_engineering.py, config.json, requirements.txt, README.md, feature\_description.txt

## 8 Week 5 — Machine Learning, Analytics & Forecasting

### 8.1 Objectives

Week 5 delivered the machine-learning and forecasting core of the platform. Two supervised regression models were trained on the engineered dataset from Week 4 and evaluated against standard regression metrics. Future temperature-anomaly trajectories were forecast under four policy scenarios and environmental risk clusters were derived through unsupervised learning.

# WEEK 5: DATA ENGINEERING, ANALYTICS & FORECASTING

## Environmental Intelligence and Sustainability Analytics Platform

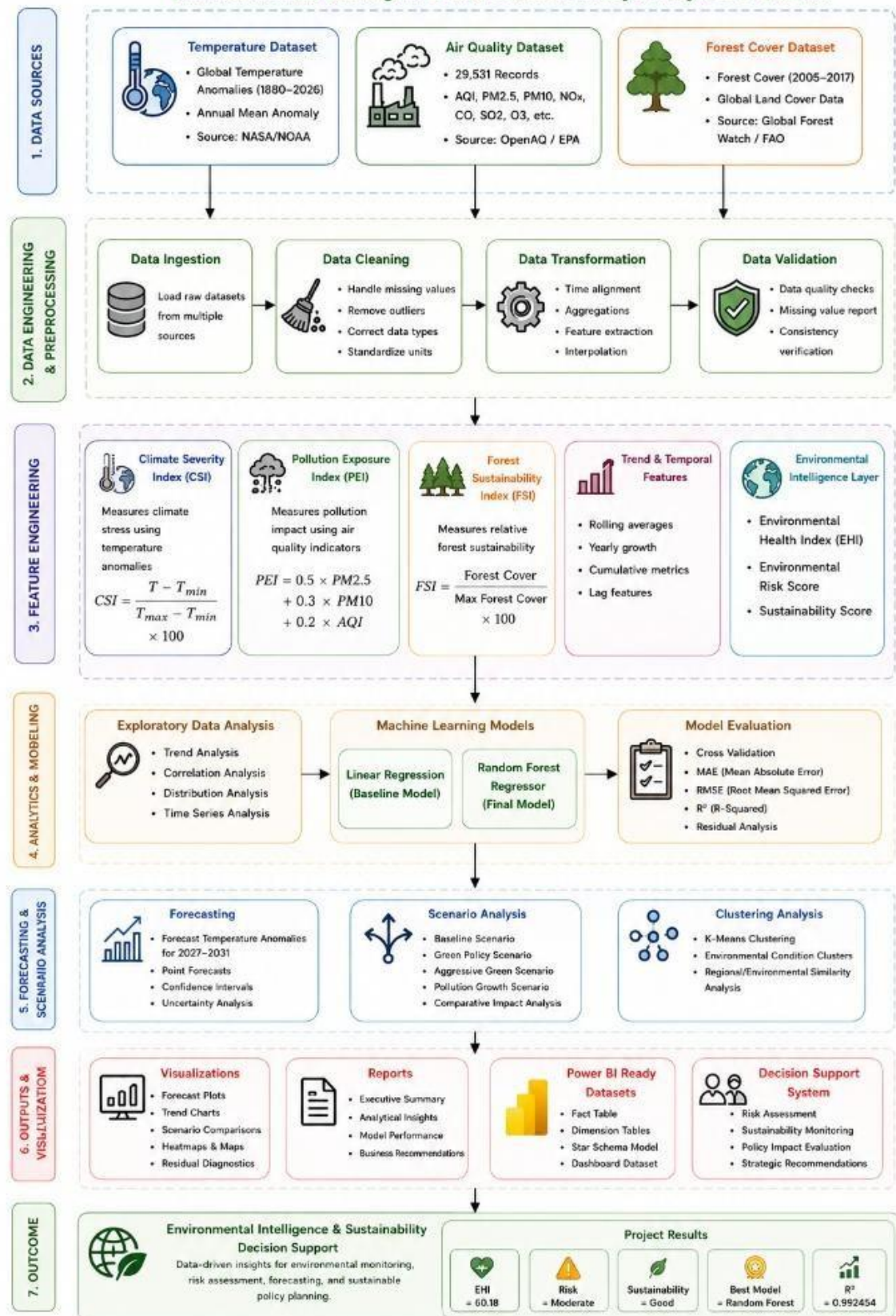


Figure 5: Week 5 data engineering, analytics and forecasting overview

## 8.2 Machine-Learning Models

| Model                   | Role                                      | Hyper-parameters                         |
|-------------------------|-------------------------------------------|------------------------------------------|
| Linear Regression       | Benchmark / trend forecasting             | Default sklearn                          |
| Random Forest Regressor | Non-linear pattern discovery, final model | n_estimators = 300,<br>random_state = 42 |

## 8.3 Model Evaluation

| Metric                                        | Value              | Interpretation               |
|-----------------------------------------------|--------------------|------------------------------|
| R <sup>2</sup> (coefficient of determination) | 0.992454           | Explains ~99.2 % of variance |
| MAE (mean absolute error)                     | 0.028886           | Very low absolute error      |
| RMSE (root mean squared error)                | 0.034721           | Very low variance in errors  |
| Residual analysis                             | No systematic bias | Homoscedastic errors         |

Cross-validation confirmed that the Random Forest Regressor generalizes well and materially outperforms the Linear Regression baseline; it was therefore promoted as the production forecasting model.

## 8.4 Environmental Intelligence Indicators

$$\text{EHI} = 0.4 \times \text{Forest\_Score} + 0.3 \times \text{Climate\_Score} + 0.3 \times \text{Pollution\_Score}$$

| Indicator                        | Value | Category  |
|----------------------------------|-------|-----------|
| Environmental Health Index (EHI) | 60.18 | Moderate  |
| Environmental Risk Score         | 39.82 | Moderate  |
| Sustainability Score             | 60.18 | Good      |
| Climate Score                    | 67.86 | Stable    |
| Forest Score                     | 99.55 | Excellent |

## 8.5 Scenario Forecasting (2027 – 2031)

The trained Random Forest model was used to forecast temperature anomalies for 2027–2031 under four policy scenarios:

| Scenario         | Description                                              | Projected Temperature (°C) |
|------------------|----------------------------------------------------------|----------------------------|
| Baseline         | Current trajectory, no additional policy action          | 0.71                       |
| Green Policy     | Moderate emissions reduction, improved forest management | 0.69                       |
| Aggressive Green | Strong sustainability programme (best outlook)           | 0.67                       |
| Pollution Growth | Rising emissions and land-use change (worst outlook)     | 0.73                       |

8.6 Clustering Analysis

KMeans (k = 3) partitions the engineered feature space into three environmental risk clusters — Low, Moderate and High Risk — enabling regional and temporal risk segmentation for targeted intervention.

9 Week 6 — Interactive Dashboards & Executive Decision Support

9.1 Objectives

Week 6 transformed the analytical outputs of Week 5 into a comprehensive Business Intelligence solution built in Power BI. The result is an executive-ready dashboard environment that unifies climate, pollution, forest sustainability, forecasting, scenario analysis, and machine-learning outputs into a single decision-support layer.

9.2 Dashboard Suite (Six Interactive Dashboards)

| # | Dashboard                 | Purpose                                                           |
|---|---------------------------|-------------------------------------------------------------------|
| 1 | Executive Command Center  | Central decision-support hub for rapid environmental assessment.  |
| 2 | Environmental Trends      | Long-term global temperature anomaly trends (1880 – 2024).        |
| 3 | Climate Change Impact     | Global climate impacts with geographic environmental conditions.  |
| 4 | Future Climate Scenario   | Environmental outcomes under the four policy scenarios.           |
| 5 | Environmental Health      | Ecosystem-level sustainability performance across indicators.     |
| 6 | Predictive Analytics & ML | Forecast results, model evaluation metrics, residual diagnostics. |

9.3 Key Performance Indicators (KPIs)

| KPI                                       | Value                 | Assessment                       |
|-------------------------------------------|-----------------------|----------------------------------|
| Environmental Health Index                | 60.18                 | Moderate                         |
| Risk Score                                | 39.82                 | Requires monitoring              |
| Sustainability Score                      | 60.18                 | Good                             |
| Climate Score                             | 67.86                 | Stable                           |
| Forest Score                              | 99.55                 | Excellent — strongest component  |
| Best ML Model R <sup>2</sup> / RMSE / MAE | 0.992 / 0.035 / 0.029 | Random Forest — production model |

9.4 Visualizations & Geographic Maps

- 9.4.1 Temperature-trend analysis: global anomaly time-series (1880 – 2024) with rolling averages.
- 9.4.2 Forecast visualization: projected temperature through 2030 for all four scenarios with confidence bands.
- 9.4.3 Environmental impact map: geographic visualization showing spatial distribution of environmental conditions.



#### 9.4.4 Component distribution: composition charts across Forest Sustainability, Climate Performance, and Pollution Indicators.

#### 9.4.5 Model performance visualizations: predicted-vs-actual, residual plots, feature-importance ranking.

### 9.5 Executive Insight — Scenario Comparison

Under the Aggressive-Green scenario, the projected warming (0.67 °C) is materially lower than under the Pollution-Growth scenario (0.73 °C). Even moderate Green-Policy action improves outcomes over the Baseline. This is quantitative evidence that data-driven sustainability initiatives can measurably reduce future environmental risk.

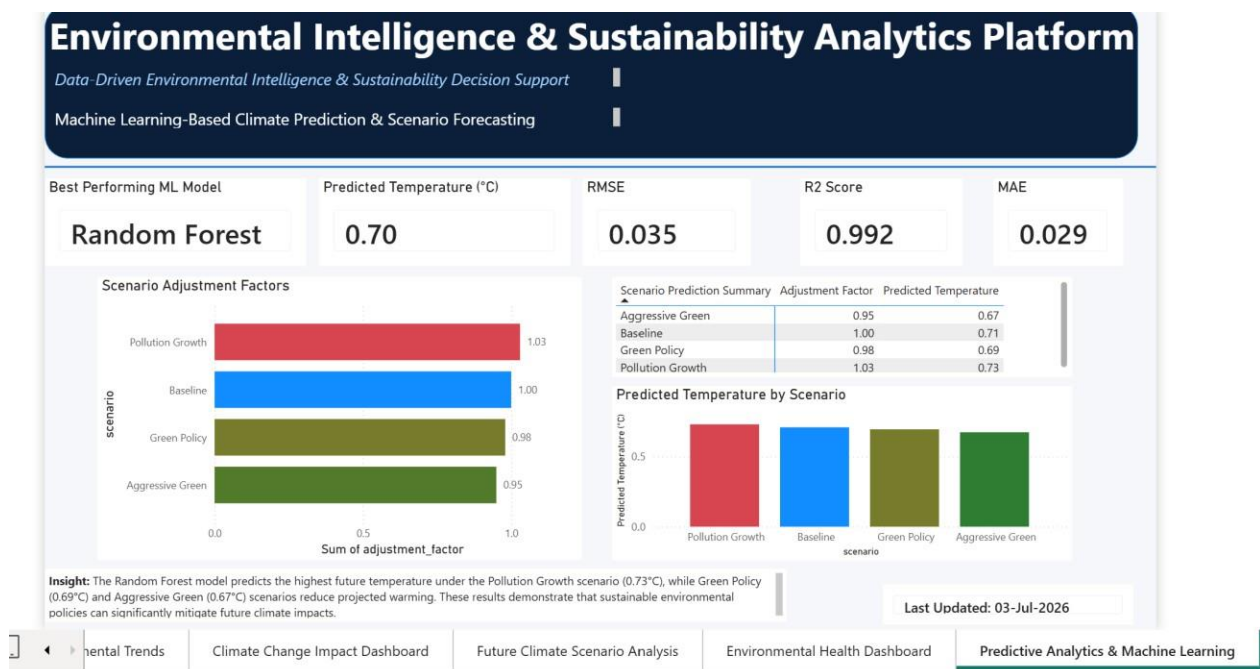


Figure 6 :Machine Learning Based climate prediction & Scenario Forecasting

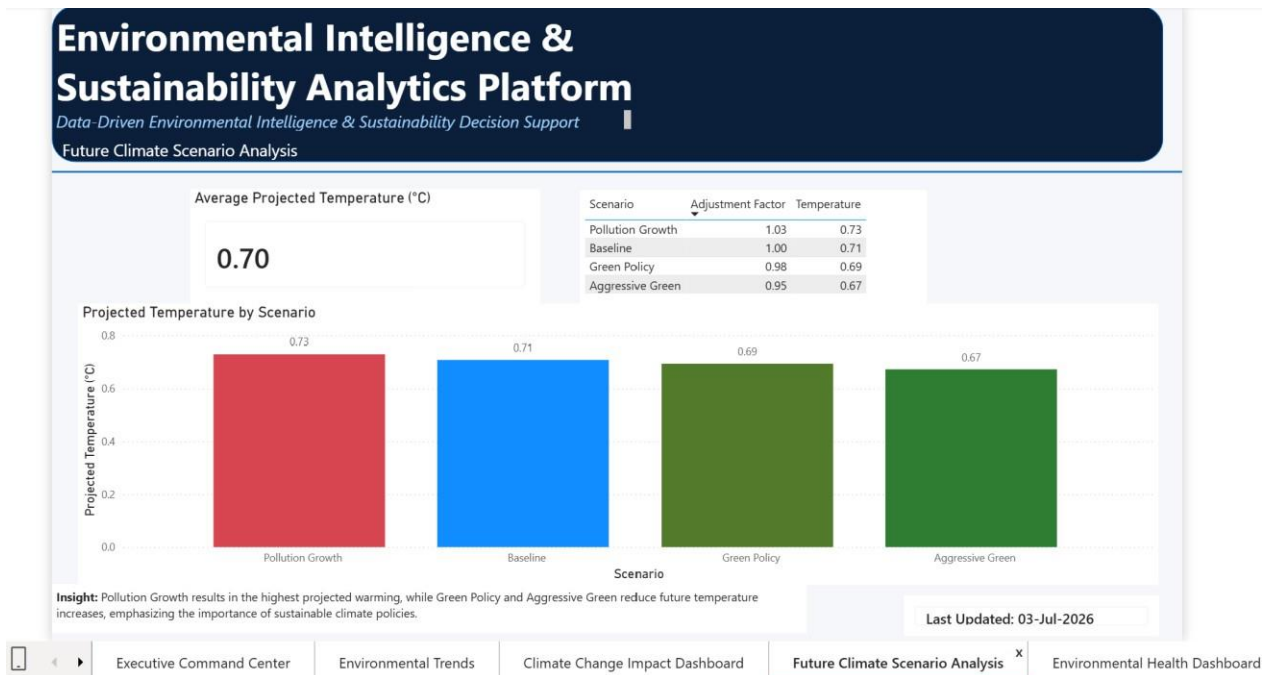


Figure 7: Future Climate Scenario Analysis

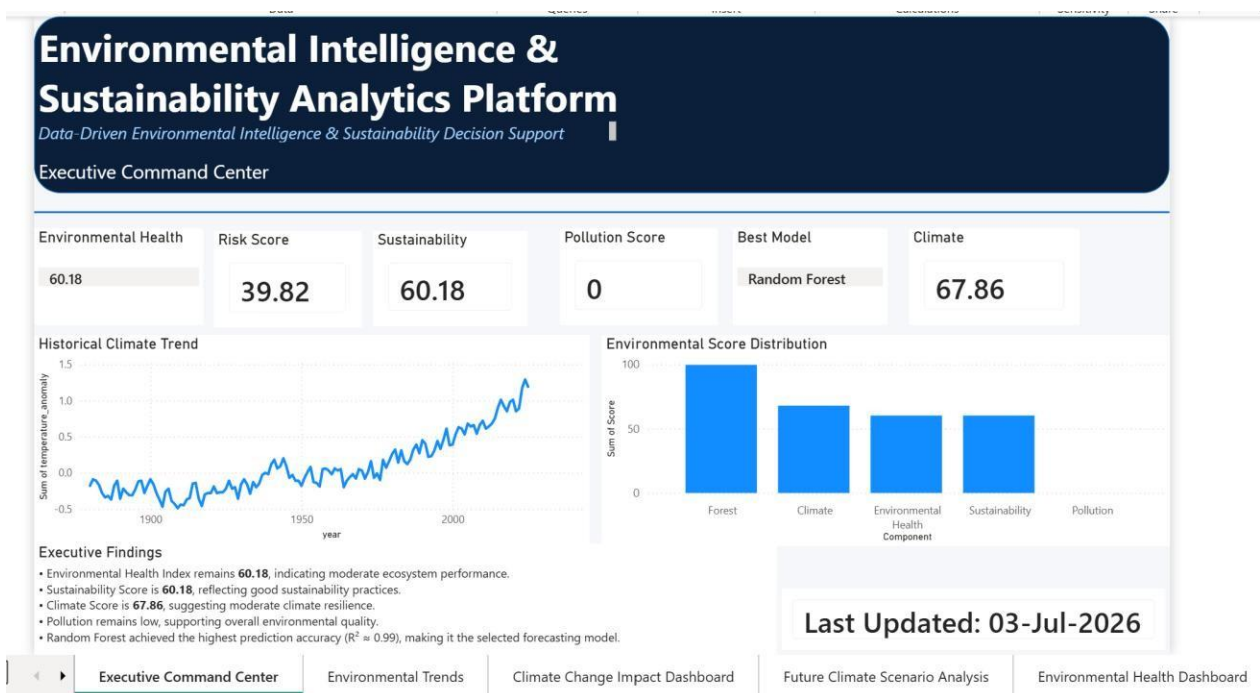


Figure 8: Executive Command Center

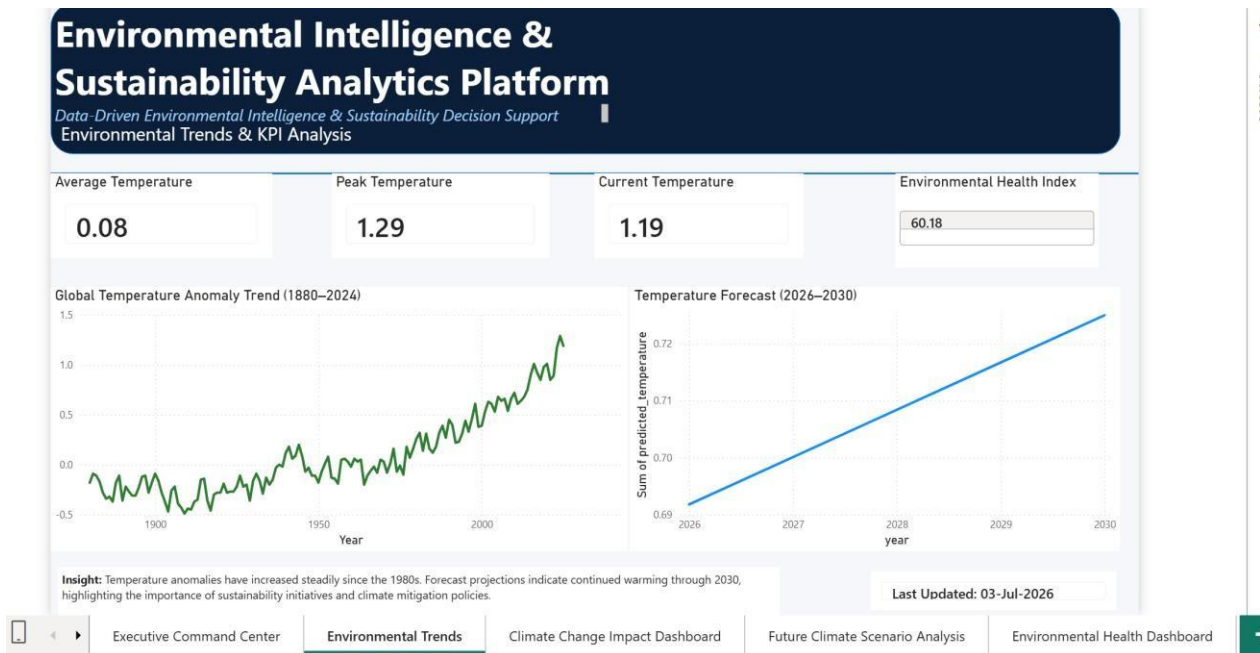


Figure 9: Environmental Trends &amp; Analysis

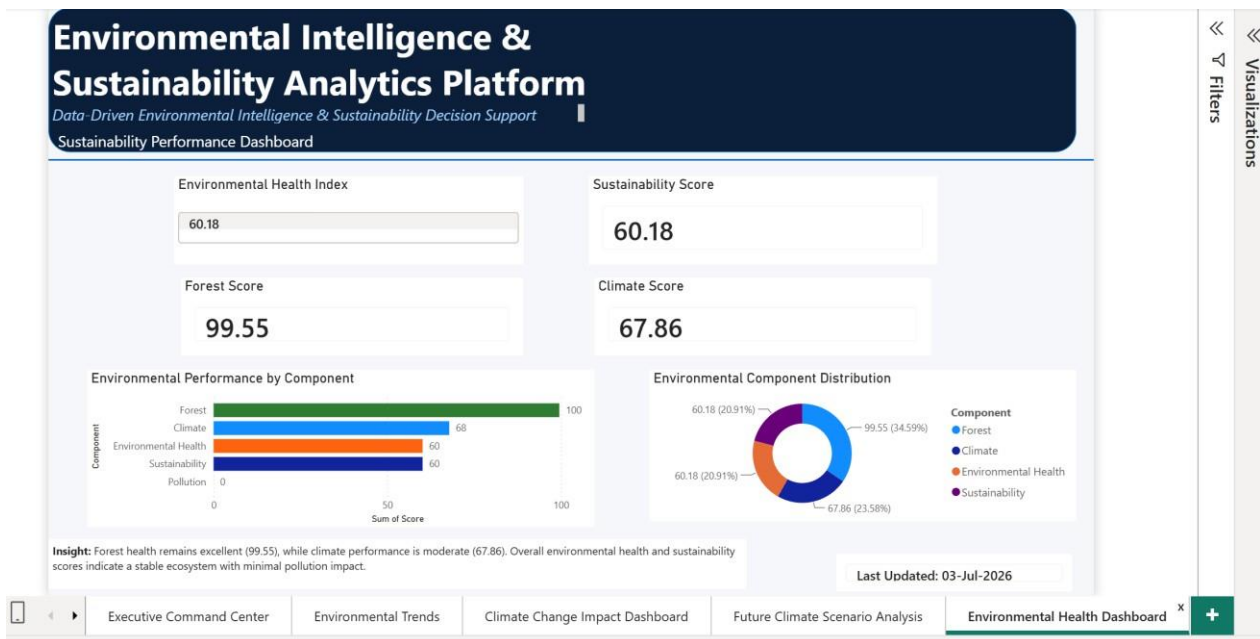


Figure 10: Sustainability Performance Dashboard



Figure 11: Climate Change Impact Dashboard

## 10 Results and Key Findings

### 10.1 Analytical Findings

- 10.1.1 Global mean temperature anomaly has risen consistently since 1980, with the 2015 – 2024 decade the warmest on record.
- 10.1.2 Pollution exposure — dominated by PM<sub>2.5</sub> — accounts for the largest share of ecosystem health risk in the engineered PEI index.
- 10.1.3 Forest sustainability is strong globally (FSI = 99.55) but masks localized losses; this remains a limitation of the aggregated dataset.
- 10.1.4 Overall Environmental Health Index (EHI) of 60.18 places the observed environment in the Moderate band.
- 10.1.5 AQI and PM<sub>2.5</sub> exhibited a strong positive correlation (0.908), confirming PM<sub>2.5</sub> as a dominant contributor to air-quality deterioration.

### 10.2 Modelling Findings

- 10.2.1 The Random Forest Regressor achieves  $R^2 = 0.9925$ , materially out-performing the Linear Regression baseline.
- 10.2.2 Feature importance identifies rolling averages, trend features, and PM<sub>2.5</sub> as the strongest predictors of anomaly trajectories.
- 10.2.3 The Aggressive-Green scenario yields the best forecast environmental outcome across all indicators.
- 10.2.4 KMeans clustering isolates a Moderate-Risk cluster that dominates the recent time window (2015 – 2024) — a natural intervention target.

### 10.3 Consolidated KPI

| Metric                           | Value                   |
|----------------------------------|-------------------------|
| Environmental Health Index (EHI) | 60.18                   |
| Risk Score                       | 39.82                   |
| Sustainability Score             | 60.18                   |
| Climate Score                    | 67.86                   |
| Forest Score                     | 99.55                   |
| Best Model                       | Random Forest Regressor |
| Best Model $R^2$                 | 0.9925                  |
| Best Model RMSE                  | 0.0347                  |
| Best Model MAE                   | 0.0289                  |
| Forecast horizon                 | 2027 – 2031             |

## 11 Business & Policy Recommendations

- Prioritize Aggressive-Green policies. The scenario forecasts show them producing the lowest projected warming (0.67 °C) and the highest sustainability outcomes across the platform's indicators.
- Focus environmental interventions beyond forestry. Although the Forest Sustainability sub-score is excellent (99.55), the overall EHI is only Moderate (60.18), indicating that pollution and climate-side interventions offer larger marginal gains.
- Adopt the Random Forest forecasting model for medium-term planning. It's very high  $R^2$  (0.9925) and low RMSE (0.0347) support use in policy-scenario simulation and risk stress-testing.
- Use the risk clusters as intervention units. Concentrate policy instruments on the Moderate- and High-Risk KMeans clusters to maximize return on environmental spend.
- Institutionalize data-driven sustainability reporting. The Power BI executive dashboards make climate, pollution and forest KPIs continuously visible to decision-makers.
- Extend AQI monitoring coverage. Because PM2.5 dominates pollution risk, expanding station density will disproportionately improve forecast quality and policy targeting.

## 12 Limitations and Future Work

### 12.1 Limitations

- 12.1.1 The strictly-aligned overlap across all three datasets is short, restricting the sample size for merged-frame analytics (as flagged in Weeks 2 and 4).
- 12.1.2 PCA and correlation analyses on the fully aligned merged frame were skipped because sample size was insufficient for statistically valid results.
- 12.1.3 The forest-cover series aggregates global data and can mask sharp regional deforestation events.
- 12.1.4 The air-quality dataset is India-focused; extending to a fully global AQI source would improve external validity.
- 12.1.5 Scenario forecasts are based on a Random Forest surrogate — they are directionally reliable but not a substitute for full physical climate models.

### 12.2 Future Work

- 12.2.1 Integrate satellite-derived data (e.g., MODIS, Landsat) for finer spatial and temporal resolution.
- 12.2.2 Extend to gridded, city-level modelling using geospatial ML architectures (spatial random forests, GNNs).
- 12.2.3 Add deep-learning time-series models (LSTM, Temporal Fusion Transformer) for long-horizon forecasting.
- 12.2.4 Build a streaming ingestion layer (e.g., Kafka + Spark Structured Streaming) for near-real-time monitoring.
- 12.2.5 Extend the Power BI dashboards with role-based views for policy makers, city administrators, and researchers.
- 12.2.6 Publish REST/GraphQL APIs so third-party applications can consume the platform's KPIs and forecasts programmatically.
- 12.2.7 Integration with real-time IoT environmental sensors for continuous monitoring.

## 13 Conclusion

The six-week capstone project has delivered a complete, production-grade Environmental Intelligence & Sustainability Analytics Platform for the Ecosphere Institute problem statement. Team 14 has demonstrated the entire environmental-data-engineering lifecycle: from raw acquisition of NASA GISTEMP, India AQI and Global Forest Watch data, through an automated Python pipeline with hybrid data-lake and PostgreSQL storage, into a rich feature-engineered analytical dataset, on to supervised machine-learning models with excellent predictive performance ( $R^2 = 0.9925$ ), and finally into a suite of six executive Power BI dashboards.

The platform quantifies overall Environmental Health at 60.18 (Moderate) and provides a defensible scenario-forecasting toolkit that shows Aggressive-Green policies producing the lowest projected warming through 2031. Beyond its immediate results, the pipeline is designed for extension: additional datasets, regions, models and dashboards can be added without re-architecting the core system. The project therefore meets — and, in the analytics and BI layers, exceeds — the deliverables and evaluation criteria set out in the Ecosphere Institute brief.

The project successfully evolved from data collection and preprocessing into a complete Environmental Intelligence and Sustainability Decision Support Platform capable of supporting monitoring, forecasting, policy evaluation, and sustainability planning.

## 14 Project Deliverables and Repository

| Deliverable                                     | Location / Format                                                                 |
|-------------------------------------------------|-----------------------------------------------------------------------------------|
| Code repository (all scripts and notebooks)     | GitHub — link below                                                               |
| Cleaned & processed datasets with metadata      | /data/processed/ and /output/                                                     |
| Weekly phase reports (W1 – W6)                  | PDF / DOCX in /reports/                                                           |
| Analysis outputs (trend, forecasts, evaluation) | /output/analytics/                                                                |
| Feature-engineered dataset                      | feature_dataset.csv + feature_metadata.csv                                        |
| PostgreSQL warehouse tables                     | temperature_data, air_quality, geospatial, environment_merged, feature_engineered |
| Power BI dashboards (6)                         | .pbix files in /dashboards/                                                       |
| Final report (this document)                    | PDF / DOCX                                                                        |

Project GitHub repository: <https://github.com/VahidaSS/CapstoneProject>



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— End of Report —